# Project Design Phase-II
# Technology Stack (Architecture & Stack)

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| Date | 15/10/2026 |
| Team ID | |
| Project Name | Metro Ticket Booking System using ServiceNow |
| Maximum Marks | 4 Marks |

## Technical Architecture:

The deliverable includes the architectural diagram below and the information as per Table-1 and Table-2.

Example: Metro ticket booking and QR ticket generation on the ServiceNow platform

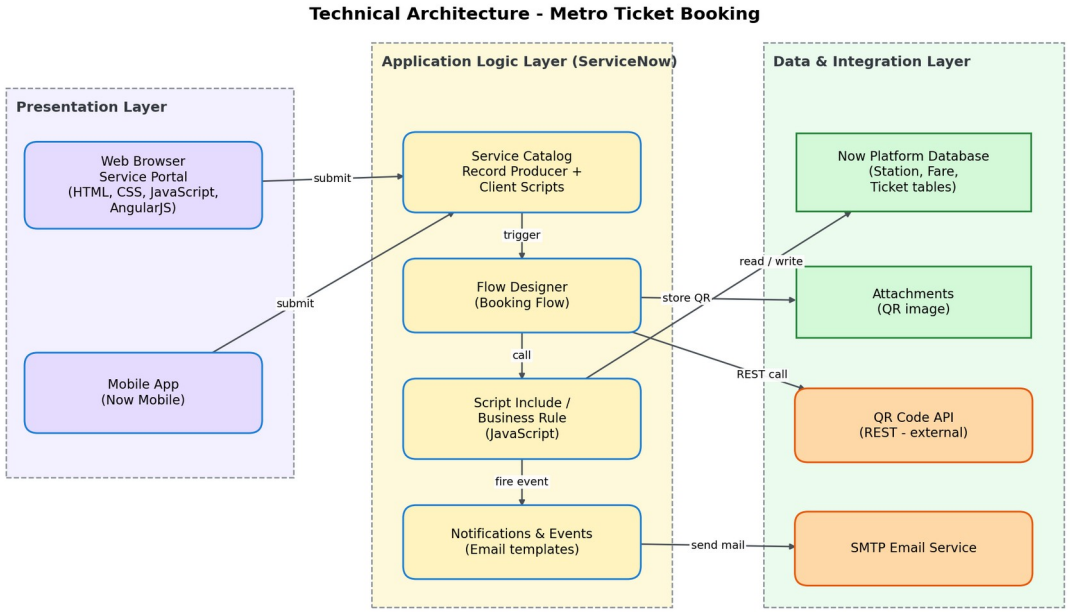


Table-1 : Components & Technologies:

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1 | User Interface | How the user interacts with the application: booking form and My Tickets page on web and mobile | Service Portal (HTML, CSS, JavaScript, AngularJS), Now Mobile |
| 2 | Application Logic-1 | Service Catalog item with variables, UI policies and client scripts for validation | ServiceNow Service Catalog / Record Producer, JavaScript |
| 3 | Application Logic-2 | Booking flow: fare lookup, ticket creation, QR generation | ServiceNow Flow Designer, Script Include (JavaScript) |
| 4 | Application Logic-3 | Email notifications for confirmation, reminder and cancellation | ServiceNow Notifications and Events |
| 5 | Database | Tables for stations, fare matrix and tickets (ticket number, stations, date, passengers, fare, status) | ServiceNow relational database (Now Platform tables) |
| 6 | Cloud Database | Database service on the cloud | ServiceNow cloud instance |
| 7 | File Storage | Storage of the QR code image for each ticket | ServiceNow Attachments |
| 8 | External API-1 | Generate the QR code image from the ticket reference | QR code generation REST API (called from Flow Designer / script) |
| 9 | External API-2 | Online payment for tickets (future scope) | Payment gateway API, e.g. Razorpay / PayU sandbox |
| 10 | Machine Learning Model | Not used in the current scope; possible future use for crowd and demand prediction | ServiceNow Predictive Intelligence (future) |
| 11 | Infrastructure (Server / Cloud) | Application development and deployment | ServiceNow cloud instance (Personal Developer Instance for development) |

Table-2: Application Characteristics:

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1 | Open-Source Frameworks | Open-source libraries used in the portal front end | AngularJS, Bootstrap |
| 2 | Security Implementations | Role-based access, ACLs on ticket tables, encrypted connections, authenticated users only | ServiceNow roles and ACLs, HTTPS/TLS, SSO (optional) |
| 3 | Scalable Architecture | Multi-tier platform architecture (presentation, application, database); new stations and fares are only data | ServiceNow Now Platform |
| 4 | Availability | Cloud-hosted instance with platform-managed redundancy and backups | ServiceNow cloud infrastructure |

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 5 | Performance | Automated asynchronous flows, indexed lookup of stations and fares, lightweight forms | Flow Designer, table indexes |

References:


- https://c4model.com/
- https://www.servicenow.com/docs
- https://developer.servicenow.com/